

Recall that this study compares the efficacy of each classifier for correctly **recalling** fraudulent credit card activity. The classifiers are tested on a data set containing only PCA (Principal Component Analysis) -transformed values (columns V1-V28) and a data set containing PCA-transformed features in addition to human readable features (amount and time). The purpose of this study is to verify whether the cost of having the data set include the human readable features is worth it. Including human readable columns sacrifices data minimization and anonymization since knowing the time of a fraud and the amount that was flagged can provide someone clues to possibly identify that transaction and the victim. Thus, justifying the inclusion of these columns can only be proven by showing that they actually provide a statistically significant advantage in the classifiers' ability to identify fraudulent activity in such a way that compensates for the reduced data minimization, anonymity, and privacy.

Below are the results for each classifier.

K Nearest Neighbour (KNN) Classifier

Full Data Set:

```
k=3 | Recall: 0.7653 | F1: 0.8475
k=5 | Recall: 0.7653 | F1: 0.8427
k=7 | Recall: 0.7857 | F1: 0.8556
k=9 | Recall: 0.7755 | F1: 0.8492
k=11 | Recall: 0.7755 | F1: 0.8492
k=13 | Recall: 0.7653 | F1: 0.8427
```

PCA Only Data Set:

```
k=3 | Recall: 0.7959 | F1: 0.8619
k=5 | Recall: 0.7857 | F1: 0.8556
k=7 | Recall: 0.7857 | F1: 0.8556
k=9 | Recall: 0.7653 | F1: 0.8427
k=11 | Recall: 0.7755 | F1: 0.8492
k=13 | Recall: 0.7755 | F1: 0.8492
```

Optimal k value (number of nearest labeled data points to help classify each test data point) is the same (k=7) for both sets. As a result, going forward, k=7 will be the parameter for future KNN classification trials.

Full Data Set:

```
--- Optimal KNN Classification Report (Full Dataset)---
      precision    recall  f1-score   support

     0       1.00      1.00      1.00     56864
     1       0.94      0.79      0.86        98

 accuracy                   1.00     56962
 macro avg       0.97      0.89      0.93     56962
 weighted avg    1.00      1.00      1.00     56962

Mean Cross Validation Recall: 0.7729
```

PCA Only Data Set:

```
--- Optimal KNN Classification Report (PCA Only Dataset)---
      precision    recall  f1-score   support

     0       1.00      1.00      1.00     56864
     1       0.94      0.79      0.86        98

 accuracy                   1.00     56962
 macro avg       0.97      0.89      0.93     56962
 weighted avg    1.00      1.00      1.00     56962

Mean Cross Validation Recall: 0.7750
```

Mean recall for the full data set is 0.7729 while the mean recall for the PCA only data set is 0.7750. This means that for both data sets, KNN correctly identifies a fraudulent transaction about 77% percent of the time.

PCA Only vs Full Data Set Comparison For KNN		
t-Test: Paired Two Sample for Means		
	Variable 1	Variable 2
Mean	0.772938776	0.774965986
Variance	0.003140831	0.002850253
Observations	30	30
Pearson Correlation	0.975583488	
Hypothesized Mean Difference	0	
df	29	
t Stat	-0.897194689	
P(T<=t) one-tail	0.188501639	
t Critical one-tail	1.699127027	
P(T<=t) two-tail	0.377003277	
t Critical two-tail	2.045229642	

Paired t-test results revealed no statistically significant difference between the two means ($p = 0.377$ at a 95% CI). This supports the study's null hypothesis. This also indicates that the inclusion of human-readable features does not provide a measurable performance advantage for KNN-based fraud detection. Consequently, financial institutions can enhance data minimization and privacy by utilizing only PCA-transformed components without compromising the model's ability to identify fraudulent activity. This supports a 'Privacy-First' architecture that reduces storage of potentially sensitive information.

Naive Bayes Classifier (NB)

Full Data Set:

```
Initial Split | Recall: 0.8163 | F1: 0.1124
--- Naive Bayes Classification Report (Full Dataset)---
      precision    recall  f1-score   support

     0       1.00      0.98      0.99     56864
     1       0.06      0.82      0.11         98

 accuracy          0.98     56962
 macro avg       0.53      0.90      0.55     56962
 weighted avg     1.00      0.98      0.99     56962

Mean Cross Validation Recall: 0.8293
```

PCA Only Data Set:

```
Naive Bayes Classifier Test (PCA Only Set)
Initial Split | Recall: 0.8163 | F1: 0.1108
--- Naive Bayes Classification Report (PCA Only Dataset)---
      precision    recall  f1-score   support

     0       1.00      0.98      0.99     56864
     1       0.06      0.82      0.11         98

 accuracy          0.98     56962
 macro avg       0.53      0.90      0.55     56962
 weighted avg     1.00      0.98      0.99     56962

Mean Cross Validation Recall: 0.8293
```

Mean recall for the full data set is 0.8293 for both the full data set and the PCA data set. As a result, NB can successfully identify nearly 83% of fraudulent cases. The identicalness across the two types of data sets means that, for the NB algorithm, the additional human readable columns do not change the class conditional probability equations for the two classes enough to change any of the classifications.

Due to the identicalness of the two mean recall values between the two data sets, no t test was required to test for a statistically significant difference. This supports the study's null hypothesis.

Logistic Regression (LR)

Full Data Set

```
Logistic Regression Classifier Test (Full Set)
--- Logistic Regression Classification Report (Full Dataset)---
              precision    recall  f1-score   support

     0       1.00      1.00      1.00     56864
     1       0.85      0.56      0.67        98

 accuracy          1.00      1.00      1.00     56962
 macro avg       0.92      0.78      0.84     56962
weighted avg       1.00      1.00      1.00     56962

Mean Cross Validation Recall: 0.6199
```

PCA Only Data Set

```
--- Logistic Regression Classification Report (PCA Only Dataset)---
              precision    recall  f1-score   support

     0       1.00      1.00      1.00     56864
     1       0.85      0.56      0.67        98

 accuracy          1.00      1.00      1.00     56962
 macro avg       0.92      0.78      0.84     56962
weighted avg       1.00      1.00      1.00     56962

Mean Cross Validation Recall: 0.6144
```

Mean recall for the full data set is 0.6199 while the mean recall for the PCA only data set is 0.6144. Consequently, LR performs the worst at identifying fraudulent transactions with a low recall rate of just over 61% for both data sets.

PCA Only vs Full Data Set Comparison For LR		
t-Test: Paired Two Sample for Means		
	Variable 1	Variable 2
Mean	0.619877551	0.61444898
Variance	0.004641056	0.005173334
Observations	30	30
Pearson Correlation	0.975332739	
Hypothesized Mean Difference	0	
df	29	
t Stat	1.857682935	
P(T<=t) one-tail	0.036697412	
t Critical one-tail	1.699127027	
P(T<=t) two-tail	0.073394823	
t Critical two-tail	2.045229642	

Similar to the paired t test analysis for the KNN classifier, for LR, the differences between the mean recall values across the two types of data sets is not statistically significant ($p = 0.0734$ at 95% CI). This supports the study's null hypothesis. Consequently, there is no advantage statistically in introducing the human readable features of amount and time. Because of this, financial institutions that utilize logistic regression would be unnecessarily lowering their data minimization, privacy, storage, and anonymity by including these features in their data store.

Linear SVC

Full Data Set

```
Linear SVC Classifier Test (Full Set)
--- Linear SVC Classification Report (Full Dataset)---
              precision    recall  f1-score   support

     0       1.00      0.98      0.99     56864
     1       0.08      0.92      0.15         98

 accuracy          0.98     56962
 macro avg       0.54      0.95      0.57     56962
 weighted avg    1.00      0.98      0.99     56962

Mean Cross Validation Recall: 0.9044
```

PCA Only Data Set

```
Linear SVC Classifier Test (PCA Only Set)
--- Linear SVC Classification Report (PCA Only Dataset)---
              precision    recall  f1-score   support

     0       1.00      0.98      0.99     56864
     1       0.08      0.91      0.15         98

 accuracy          0.98     56962
 macro avg       0.54      0.95      0.57     56962
 weighted avg    1.00      0.98      0.99     56962

Mean Cross Validation Recall: 0.9030
```

Mean recall for the full data set is 0.9044 while the mean recall for the PCA only data set is 0.9030. Accordingly, linear SVC is able to correctly spot over 90% of fraudulent cases across both data sets.

PCA Only vs Full Data Set Comparison For Linear SVC		
t-Test: Paired Two Sample for Means		
	Variable 1	Variable 2
Mean	0.904408163	0.903047619
Variance	0.001554104	0.001570343
Observations	30	30
Pearson Correlation	0.982239746	
Hypothesized Mean Difference	0	
df	29	
t Stat	1	
P(T<=t) one-tail	0.162790994	
tCritical one-tail	1.699127027	
P(T<=t) two-tail	0.325581988	
tCritical two-tail	2.045229642	

As with the earlier paired t test evaluations, the high p value (above 0.05) indicates that, similarly for KNN, NB, LR, Linear SVC machine learning algorithms do not gain from including the human readable features in their analysis ($p = 0.326$ at 95% CI). This supports the study's null hypothesis. As a result, this is another example in which including these features unnecessarily compromises data minimization and autonomy, leading to potential privacy issues if the amount and time features contribute to data linking used to identify a transaction. Therefore, it is not advisable to include these unnecessary features that provide no statistically significant advantage to the classifier algorithm.

In all four cases where the classifier's performance for both data sets were compared, there was found to be no statistical significance in the recall values. As a result, the study's null hypothesis was substantiated.

The comparative analysis of each classifier's recall between the two data sets indicates that inclusion of the amount and time human readable features merely reduces the data minimization and anonymity of each transaction rather than providing a statistical edge for the classifier's recall. Therefore, the results suggest that Time and Amount do not provide additional discriminative power beyond the PCA-transformed features.

Comparative Analysis

Comparing the efficacy of the various classifiers when applied on the full data set and on the PCA only data set is useful in evaluating which one is the best in which context. As mentioned earlier, while previous research has done similar comparative analyses across classifier algorithms to identify the one best at identifying fraud, none of them verified if the differences they observed were statistically significant by conducting a pair t test. Doing so allows one to gauge if the differences in recall between the two classifiers is functional or just due to random sampling errors.

PCA Only Set KNN vs NB			Full Set KNN vs NB		
t-Test: Paired Two Sample for Means			t-Test: Paired Two Sample for Means		
	Variable 1	Variable 2		Variable 1	Variable 2
Mean	0.774965986	0.829347	Mean	0.772938776	0.829347
Variance	0.002850253	0.002245	Variance	0.003140831	0.002245
Observations	30	30	Observations	30	30
Pearson Correlation	0.745980057		Pearson Correlation	0.736402584	
Hypothesized Mean Difference	0		Hypothesized Mean Difference	0	
df	29		df	29	
t Stat	-8.194392324		t Stat	-8.044735444	
P(T<=t) one-tail	2.45577E-09		P(T<=t) one-tail	3.578E-09	
t Critical one-tail	1.699127027		t Critical one-tail	1.699127027	
P(T<=t) two-tail	4.91155E-09		P(T<=t) two-tail	7.156E-09	
t Critical two-tail	2.045229642		t Critical two-tail	2.045229642	

The analysis shows that, in the case of both the PCA only data set and the full data set, the NB classifier outperformed the KNN classifier by a statistically significant margin ($p < 0.05$ for both

data sets). This contradicts the null hypothesis, which substantiates the study's hypothesis that a clear hierarchy of classifiers will be demonstrated. This also suggests that, for credit card transaction data sets similar to the one used in this study, agencies and financial institutions should choose NB over KNN regardless of if they include human-readable features or not. As shown by the paired t test results, NB, in addition to being much more computationally friendly than KNN, is also significantly more accurate.

PCA Only Set KNN vs LR			Full Set KNN vs LR		
t-Test: Paired Two Sample for Means			t-Test: Paired Two Sample for Means		
	Variable 1	Variable 2		Variable 1	Variable 2
Mean	0.774965986	0.614449	Mean	0.772938776	0.619878
Variance	0.002850253	0.005173	Variance	0.003140831	0.004641
Observations	30	30	Observations	30	30
Pearson Correlation	0.611324848		Pearson Correlation	0.645285538	
Hypothesized Mean Difference	0		Hypothesized Mean Difference	0	
df	29		df	29	
t Stat	15.23867035		t Stat	15.69124017	
P(T<=t) one-tail	1.11696E-15		P(T<=t) one-tail	5.22439E-16	
t Critical one-tail	1.699127027		t Critical one-tail	1.699127027	
P(T<=t) two-tail	2.23391E-15		P(T<=t) two-tail	1.04488E-15	
t Critical two-tail	2.045229642		t Critical two-tail	2.045229642	

The analysis between KNN and LR shows that not only does LR perform the worst at identifying fraudulent transactions, but that the margin is statistically significant and not due to chance ($p < 0.05$ for both data sets). This contradicts the null hypothesis, which substantiates the study's hypothesis that a clear hierarchy of classifiers will be demonstrated. This also highlights an innate inferiority of this classifier compared to the others, as it is statistically significantly lower than the classifier with the second lowest recall score. The low LR score compared to the rest also demonstrates LR's inability to be reliable for very unbalanced datasets such as the one used in this study with nearly 300 thousand total instances yet less than 500 fraudulent instances.

PCA Only Set Linear SVC vs NB			Full Set Linear SVC vs NB		
t-Test: Paired Two Sample for Means			t-Test: Paired Two Sample for Means		
	Variable 1	Variable 2		Variable 1	Variable 2
Mean	0.903047619	0.829347	Mean	0.904408163	0.829347
Variance	0.001570343	0.002245	Variance	0.001554104	0.002245
Observations	30	30	Observations	30	30
Pearson Correlation	0.60228459		Pearson Correlation	0.595612422	
Hypothesized Mean Difference	0		Hypothesized Mean Difference	0	
df	29		df	29	
t Stat	10.24105239		t Stat	10.36225856	
P(T<=t) one-tail	1.91697E-11		P(T<=t) one-tail	1.46294E-11	
t Critical one-tail	1.699127027		t Critical one-tail	1.699127027	
P(T<=t) two-tail	3.83395E-11		P(T<=t) two-tail	2.92587E-11	
t Critical two-tail	2.045229642		t Critical two-tail	2.045229642	

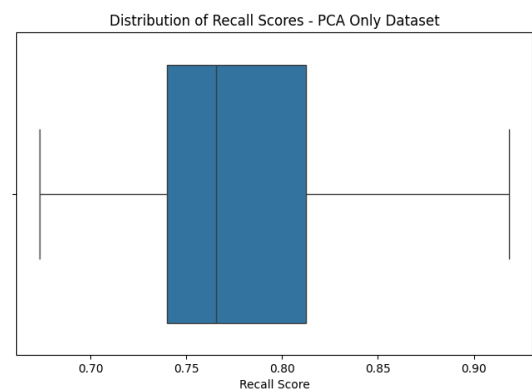
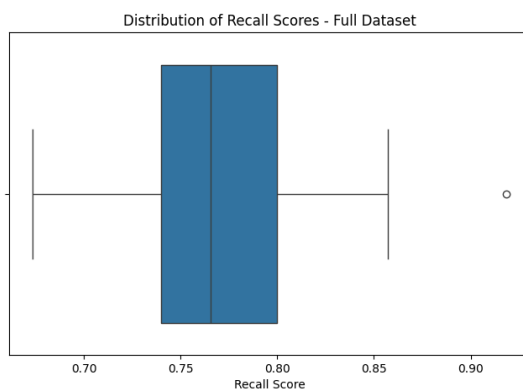
The analysis shows that linear SVC is the best classifier for both the PCA only and full data sets by a statistically significant margin ($p < 0.05$ for both data sets). This contradicts the null hypothesis, which substantiates the study's hypothesis that a clear hierarchy of classifiers will be

demonstrated. As a result, it is, according to this study, the optimal classifier to use when trying to identify fraudulent transactions. Linear SVC, by nature of its large array of hyperparameters, also presents opportunities for tinkering with the model further to see if it could yield even better results. This also indicates how, despite both linear SVC and logistic regression relying on linearity, linear SVC's ability to maximize margin using support vectors and allows it to handle imbalanced data effectively.

The comparative analysis substantiated the study's hypothesis which was that, for each paired t test, the null hypothesis will be false, resulting in a clear, demonstrable hierarchy of efficacy of classifier. For both cases that hierarchy was linear SVC as the best, followed by NB, then KNN, then LR as the worst.

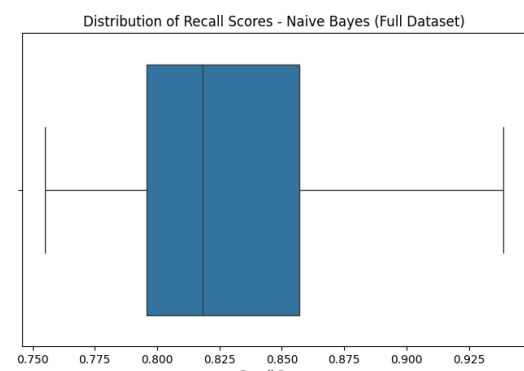
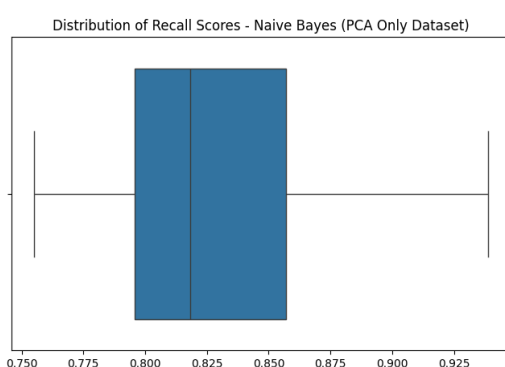
Stability of Each Classifier

KNN



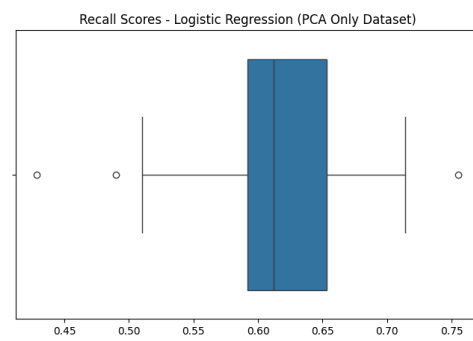
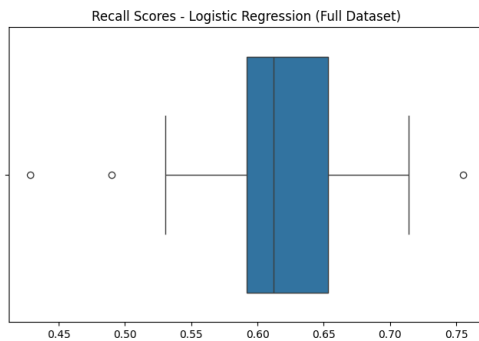
The above box charts demonstrate that the stability of the KNN classifier did not change much when the human-readable features were removed. In fact, removing those two features eliminated an outlier in the classification process and instead incorporated that point into its fourth quartile. The spread of the middle 50% of the data in both cases are very similar, though very slightly wider for the PCA set. The overall spread is larger for the PCA only set. This means that, while the PCA only data set eliminates an outlier, it has a slightly more dramatic range of possible recall scores, making it a little bit less predictable than the full data set.

NB



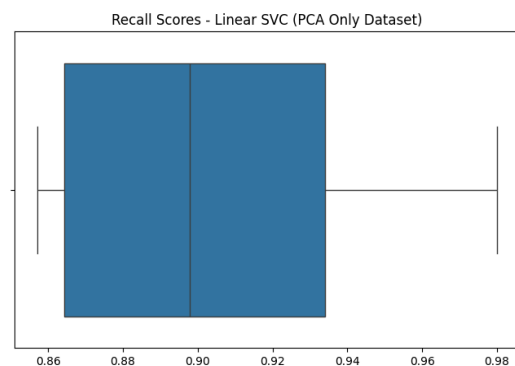
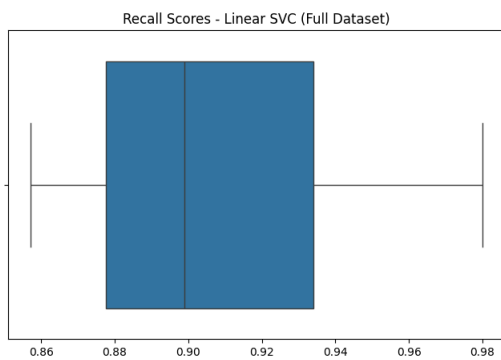
Unsurprisingly, the identicalness of the recall values for the NB classifier, caused by the removal of the human readable features not affecting the class conditional probabilities enough to actually change classifications of instance data, resulted in identical box charts with the same spreads. This indicates that, not only is there no advantage of having the full data set with human-readable features in terms of average recall, but including it does not even stabilize the predictions if NB is being used.

LR



Similarly with NB, for LR the box charts for both data sets are very similar if not identical. Unlike the KNN box chart, where the PCA only classification eliminated an outlier and incorporated that into its regular spread, all outliers were preserved across both data sets, along with the spread of the middle 50% of the data and the overall spread of standard points being very similar. This suggests that the human-readable features incorporated into the data set do not even provide slightly more predictability for an LR classification system.

Linear SVC



The Linear SVC is where noticeable differences between the box charts between the full data set and the PCA only data set arise. Although both graphs have similar overall spreads, the PCA only data set has a larger spread for its middle 50% of data points. Interestingly, this also shifts that spread down, demonstrating the greater volatility of the PCA only data set for its middle 50% and how it allows for lower recall scores amongst its middle 50%. This demonstrates how in terms of predictability, the full data set is slightly superior to the PCA only data set, along with having a slightly superior middle 50% in its recall scores.